

14 c) one-to-one and onto.

a) Solution:

[S] d) neither one-to-one nor onto.

Solution: $f(x) = x^2$. $f(-2) = (-2)^2 = 4$ $f(2) = 2^2 = 4$. But $2 \neq -2$

By Defn. 5, f is not one-to-one.

Let, $f(x) = y$ for some $y \in \mathbb{Z}$. $\therefore x^2 = y \Rightarrow x = \pm\sqrt{y}$.

If y is not a perfect square, then $\pm\sqrt{y} \notin \mathbb{Z}$.

$\therefore \nexists y$ s.t. $\forall x, y \in \mathbb{Z}^+$ and $x \in \mathbb{Z}$, $f(x) = y$.

By Defn. 7, f is not onto $\square$

22) Determine whether each of these functions is a bijection from $\mathbb{R}$ to $\mathbb{R}$.

a) $f(x) = -3x + 4$

Solution: Let, $x, y \in \mathbb{R}$ and $f(x) = f(y)$. $\therefore -3x + 4 = -3y + 4 \Rightarrow -3x = -3y$

$\Rightarrow x = y$. By Defn. 5, f is one-to-one.

Let $f(x) = y$ for some $y \in \mathbb{R}$. $\therefore -3x + 4 = y \Rightarrow x = \frac{y-4}{-3}$

It's easy to see that $\frac{y-4}{-3} \in \mathbb{R}$.

$\therefore \forall y \in \mathbb{R}, \exists x \in \mathbb{R}$ s.t. $f(x) = y$. By Defn. 7, f is onto.

By Defn. 8, f is a bijection. $\square$

b) $f(x) = -3x^2 + 7$.

Solution: We know that -1 and $1 \in \mathbb{R}$.

$f(-1) = -3(-1)^2 + 7 = -3 + 7 = 4$ $f(1) = -3(1)^2 + 7 = -3 + 7 = 4$.

But $-1 \neq 1$. By Defn. 5, f is not one-to-one.

By Defn. 8, f is not a bijection. $\square$

c) $f(x) = (x+1)/(x+2)$

Solution: The function is not defined at $x = -2$.

Let, $x, y \in \mathbb{R}$ and $f(x) = f(y)$. $\therefore \frac{x+1}{x+2} = \frac{y+1}{y+2} \Rightarrow (x+1)(y+2) = (y+1)(x+2)$